

## 2-11 Fluently Subtract within 1,000

**Learning goal:** I can subtract 3-digit numbers by decomposing, adjusting, or using a related addition equation, and choose a strategy that is efficient for me.

**Textbook:** pages 75–78. The book solves  $543 - 134$  three ways: decompose, adjust (+6 to both), and a related addition equation.

### Words to Know

|                          |                                                                                                         |
|--------------------------|---------------------------------------------------------------------------------------------------------|
| <b>difference</b>        | The answer when you subtract.                                                                           |
| <b>decompose</b>         | Break the number you are taking away into parts, then take away one part at a time.                     |
| <b>adjust</b>            | Add the same amount to both numbers (or take the same amount from both). The difference stays the same. |
| <b>related addition</b>  | Write subtraction as addition with a missing part: $564 - 145 = ? \rightarrow 145 + ? = 564$ .          |
| <b>fewer</b>             | Less. “134 fewer than 543” means $543 - 134$ .                                                          |
| <b>extra information</b> | A number in a story that you do not need.                                                               |

### How to Teach It

- Decompose:** take away the hundreds, then the tens, then the ones. Split the ones to land on a ten if that helps ( $-4$  then  $-1$ ).
- Adjust:** if the number you take away is close to a ten or hundred, add the same amount to **both** numbers (subtraction uses the same move on both).
- Related addition:** count up from the smaller number to the larger one. Add the jumps.
- Any correct strategy is fine. Ask which one is fastest for these numbers.
- In stories, find the two numbers being compared. Cross out the extra one.

**Tip:** Remember the rule: adding  $\rightarrow$  opposite moves; subtracting  $\rightarrow$  same moves. Ask, “If you change one number, what must you do to the other?”

### Worked Examples

#### Example 1: Decompose one number

$564 - 100 = 464$   
 $464 - 40 = 424$   
 $424 - 4 = 420$   
 $420 - 1 = 419$  So  $564 - 145 = \mathbf{419}$ .

#### Example 2: Adjust both numbers

$564 - 145 = ?$   
 $\begin{array}{r} +5 \\ 564 \\ \hline 569 \end{array} \quad \begin{array}{r} +5 \\ 145 \\ \hline 150 \end{array} = \mathbf{419}$

Both move up 5, so the difference stays the same.

#### Example 3: Related addition equation

$145 + ? = 564$   
 Count up:  $145 \rightarrow 150 (+5) \rightarrow 200 (+50) \rightarrow 564 (+364)$ .  
 $5 + 50 + 364 = \mathbf{419}$

#### Example 4: Find the useful numbers

The library lent 482 books in March, 129 fewer in April, and 350 in May. How many in April?  
 May is extra.  $482 - 129 \rightarrow 483 - 130 = \mathbf{353}$ .

### Skills Checklist *Check off each skill your child can do without help.*

- |                                                                         |                                                                               |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <input type="checkbox"/> Subtract by decomposing the number taken away. | <input type="checkbox"/> Choose an efficient strategy and explain it.         |
| <input type="checkbox"/> Adjust both numbers the same way.              | <input type="checkbox"/> Spot a strategy that does not work.                  |
| <input type="checkbox"/> Use a related addition equation and count up.  | <input type="checkbox"/> Ignore extra information and solve two-step stories. |

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Name \_\_\_\_\_ Date \_\_\_\_\_

**Together** *Work with your grown-up. Talk about each step.*

- A.** Find  $652 - 118$  two ways.

**Decompose 118**

**Adjust both numbers**

Difference: \_\_\_\_\_

- B.** Nina and Theo solve  $182 - 147$ . Nina writes  $180 - 145 = ?$ . Theo writes  $147 + ? = 182$ . Do both strategies work? Which would you choose? Solve.

Difference: \_\_\_\_\_

- C.** A road sign shows how far away three towns are. How much farther is Oak Hill than Lakeview?

| Town      | Miles |
|-----------|-------|
| Lakeview  | 147   |
| Pine City | 215   |
| Oak Hill  | 384   |

Miles farther: \_\_\_\_\_

**On My Own** *Try these by yourself. Show your work.*

- 1.** Solve. Use any strategy.

$$878 - 435 = \underline{\hspace{2cm}}$$

- 2.** Solve. Use any strategy.

$$652 - 286 = \underline{\hspace{2cm}}$$

3. Solve. Use any strategy.  
 $824 - 768 =$  \_\_\_\_\_

4. Solve. Use any strategy.  
 $152 - 78 =$  \_\_\_\_\_

- 
5. 713 people want tickets to a show. The theater has 227 fewer seats than that. How many seats does the theater have?

6. Sam scored 318 points. That was 105 more than his last game. Ava scored 503 points. How many more points did Ava score than Sam?

Seats: \_\_\_\_\_

More points: \_\_\_\_\_

- 
7. Solve. Explain your strategy.  
 $500 - 268 =$  \_\_\_\_\_

8. Solve. Explain your strategy.  
 $775 - 325 =$  \_\_\_\_\_

Strategy: \_\_\_\_\_

Strategy: \_\_\_\_\_

- 
9. Jay says that to find  $368 - 125$ , he can decompose 368 and subtract the parts from 125. Do you agree? Explain.

\_\_\_\_\_  
\_\_\_\_\_

Difference: \_\_\_\_\_

- 
10. Mia has 600 points. She spends 284 points on a hat and 197 points on shoes. Does she have more than 100 points left? Show how you know.

Answer: \_\_\_\_\_

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### Together

|          |                                                                                                                                                                                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A</b> | Decompose: $652 - 100 = 552$ ; $552 - 10 = 542$ ; $542 - 8 = 534$ .<br>Adjust: +2 to both $\rightarrow 654 - 120 = 534$ . Difference: <b>534</b><br><i>Also accept <math>542 - 2 = 540</math>, <math>540 - 6 = 534</math> for the ones step.</i>        |
| <b>B</b> | <b>Both work.</b> Nina took 2 from both numbers (same moves). Theo used a related addition. $182 - 147 = 35$ .<br><i>Theo: <math>147 \rightarrow 150 (+3) \rightarrow 180 (+30) \rightarrow 182 (+2) = 35</math>. Any choice with a reason is fine.</i> |
| <b>C</b> | $384 - 147 = 237$ miles farther<br><i>Pine City (215) is not needed. e.g., <math>387 - 150 = 237</math>.</i>                                                                                                                                            |

### On My Own

|           |                                                                                                                                                                                       |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b>  | <b>443</b><br><i>Items 1–4: accept any correct strategy.</i>                                                                                                                          |
| <b>2</b>  | <b>366</b><br><i>e.g., <math>656 - 290 = 366</math> (+4 to both).</i>                                                                                                                 |
| <b>3</b>  | <b>56</b><br><i>Counting up is quick: <math>768 \rightarrow 770 (+2) \rightarrow 800 (+30) \rightarrow 824 (+24) = 56</math>.</i>                                                     |
| <b>4</b>  | <b>74</b><br><i>e.g., <math>154 - 80 = 74</math>. A common mistake is 126 (taking smaller digits from larger in each place).</i>                                                      |
| <b>5</b>  | $713 - 227 = 486$ seats<br><i>"227 fewer than 713" means <math>713 - 227</math>.</i>                                                                                                  |
| <b>6</b>  | $503 - 318 = 185$ more points<br><i>The 105 is extra information.</i>                                                                                                                 |
| <b>7</b>  | <b>232</b><br><i>Sample: related addition <math>268 + ? = 500</math> (<math>2 + 30 + 200</math>), or <math>499 - 267 = 232</math>.</i>                                                |
| <b>8</b>  | <b>450</b><br><i>Sample: <math>775 - 300 = 475</math>; <math>475 - 25 = 450</math>, or adjust to <math>750 - 300</math>.</i>                                                          |
| <b>9</b>  | <b>No.</b> You must take away 125 from 368, so decompose 125 (100, 20, 5): 268, 248, <b>243</b> .<br><i>Taking parts of 368 away from 125 does not work: 125 is smaller than 368.</i> |
| <b>10</b> | $600 - 284 = 316$ ; $316 - 197 = 119$ . <b>Yes</b> , 119 is more than 100.                                                                                                            |